SOFTWARE ENGINEERING

Lab 6: Code and Branch Coverage Analysis

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Objective

The objective of this lab is to test the loan eligibility application using code coverage and branch coverage techniques. These techniques help check whether the application code and its decision-making conditions are properly tested.

1. Code Coverage Analysis

Code coverage measures how much of the application code is executed when the test cases run. In this lab, I used Python's coverage tool to measure code coverage.

I created test cases for the loan eligibility application and executed them using pytest. The coverage report showed **100% code coverage**, which means all executable lines in the measured files were executed during testing.

OUTPUT:

```
PS C:\Users\DELL\Downloads\L6_L7_26_loan_eligibility\Lab 6> py -m coverage erase
PS C:\Users\DELL\Downloads\L6_L7_26_loan_eligibility\Lab 6> py -m coverage run -m pytest test_loan_eligibility.py
==================== test session starts ====================
platform win32 -- Python 3.14.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\DELL\Downloads\L6_L7_26_loan_eligibility\Lab 6
plugins: anyio-4.15.1, platformdirs-4.13.0
collected 25 items

test_loan_eligibility.py .........................       [100%]

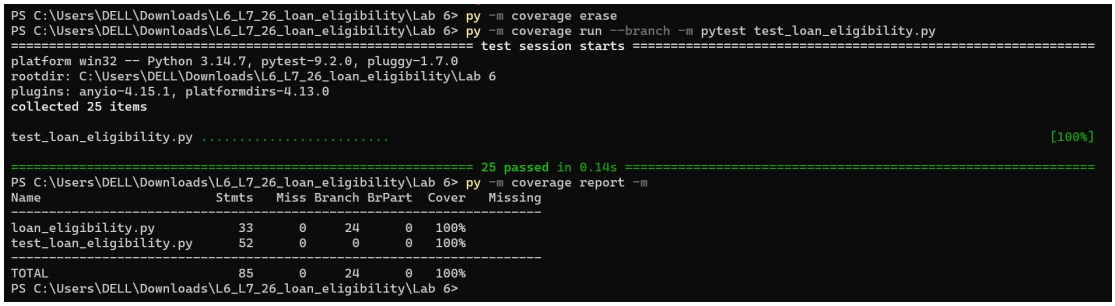
==================== 25 passed in 0.20s =====================
PS C:\Users\DELL\Downloads\L6_L7_26_loan_eligibility\Lab 6> py -m coverage report -m
Name                       Stmts   Miss  Cover   Missing
--------------------------------------------------------
loan_eligibility.py           33      0   100%
test_loan_eligibility.py      52      0   100%
--------------------------------------------------------
TOTAL                         85      0   100%
PS C:\Users\DELL\Downloads\L6_L7_26_loan_eligibility\Lab 6>
```

2. Branch Coverage Analysis

Branch coverage checks whether the different possible outcomes of decision conditions are tested. For example, the loan eligibility application checks the applicant's age, income, employment type, and credit score.

I used branch coverage to test different outcomes of these conditions, including eligible and ineligible applicants. The coverage report showed **100% branch coverage** for the measured files, indicating that all reported branches were executed by the tests.

OUTPUT:



Conclusion

In this lab, I implemented test cases for code coverage and branch coverage analysis of a loan eligibility application. The coverage reports showed 100% coverage for the measured files. This exercise helped me understand how testing techniques can identify untested code paths and improve confidence in application behaviour.

GITLINK:

1. https://github.com/krshashwath2-spec/L6_L7_26_loan_eligibility/tree/feature/loan-eligibility-coverage

2.https://github.com/krshashwath2-spec/SE_LAB_SEC_H